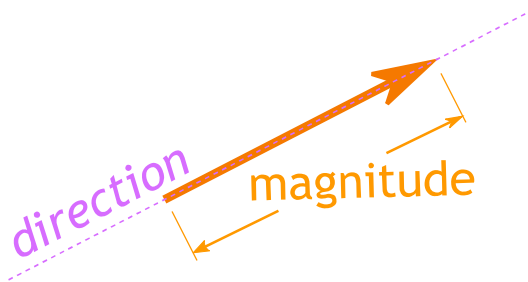


Cross Product

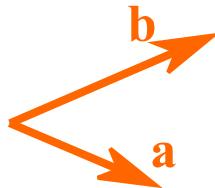
We may use [Cookies](#)

OK

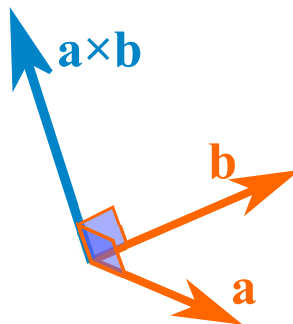
A [vector](#) has **magnitude** (how long it is) and **direction**:



Two vectors can be **multiplied** using the "**Cross Product**" (*also see* [Dot Product](#))

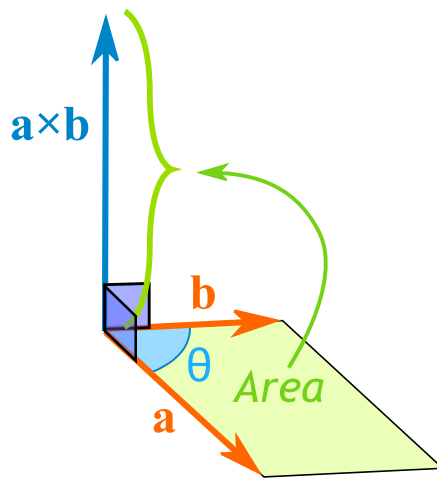


The Cross Product $\mathbf{a} \times \mathbf{b}$ of two vectors is **another vector** that is at right angles to both:

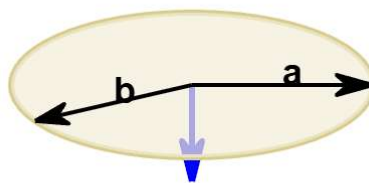


And it all happens in 3 dimensions!

The magnitude (length) of the cross product equals the [area of a parallelogram](#) with vectors $\mathbf{a}$ and $\mathbf{b}$ for sides:



See how it changes for different angles:



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The cross product (*blue*) is:

zero in length when vectors **a** and **b** point in the same, or opposite, direction

reaches maximum length when vectors **a** and **b** are at right angles

And it can point one way or the other!

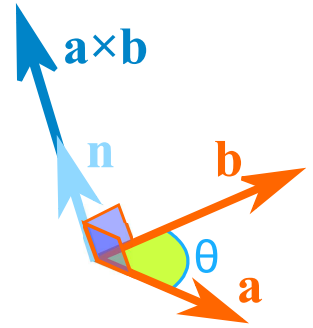
So how do we calculate it?

Calculating

WE CAN CALCULATE THE CROSS PRODUCT THIS WAY:

$$\mathbf{a} \times \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \sin(\theta) \mathbf{n}$$

- $|\mathbf{a}|$ is the magnitude (length) of vector $\mathbf{a}$
- $|\mathbf{b}|$ is the magnitude (length) of vector $\mathbf{b}$
- θ is the angle between $\mathbf{a}$ and $\mathbf{b}$
- $\mathbf{n}$ is the unit vector at right angles to both $\mathbf{a}$ and $\mathbf{b}$



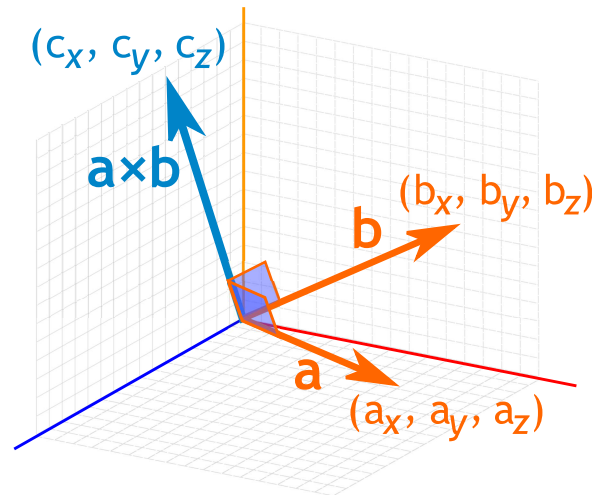
So the **length** is: the length of $\mathbf{a}$ times the length of $\mathbf{b}$ times the sine of the angle between $\mathbf{a}$ and $\mathbf{b}$,

Then we multiply by the vector $\mathbf{n}$ so it heads in the correct **direction** (at right angles to both $\mathbf{a}$ and $\mathbf{b}$).

OR WE CAN CALCULATE IT THIS WAY:

When $\mathbf{a}$ and $\mathbf{b}$ start at the origin point (0,0,0), the Cross Product will end at:

- $c_x = a_y b_z - a_z b_y$
- $c_y = a_z b_x - a_x b_z$
- $c_z = a_x b_y - a_y b_x$



Example: The cross product of $\mathbf{a} = (2,3,4)$ and $\mathbf{b} = (5,6,7)$

- $c_x = a_y b_z - a_z b_y = 3 \times 7 - 4 \times 6 = -3$
- $c_y = a_z b_x - a_x b_z = 4 \times 5 - 2 \times 7 = 6$
- $c_z = a_x b_y - a_y b_x = 2 \times 6 - 3 \times 5 = -3$

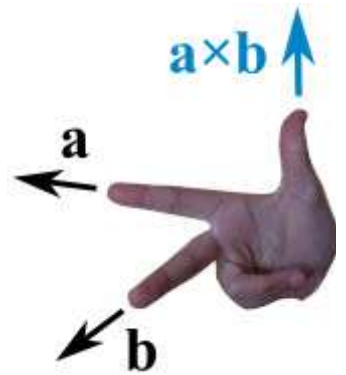
Answer: $\mathbf{a} \times \mathbf{b} = (-3, 6, -3)$

Which Direction?

The cross product could point in the completely opposite direction and still be at right angles to the two other vectors, so we have the:

"Right Hand Rule"

With your right-hand, point your index finger along vector **a**, and point your middle finger along vector **b**: the cross product goes in the direction of your thumb.



Dot Product

The Cross Product gives a **vector** answer, and is sometimes called the **vector product**.

But there is also the Dot Product which gives a **scalar** (ordinary number) answer, and is sometimes called the **scalar product**.



Question: What do you get when you cross an elephant with a banana?

Answer: $|\mathbf{elephant}| |\mathbf{banana}| \sin(\theta) \mathbf{n}$

Mathopolis: [Q1](#) [Q2](#) [Q3](#) [Q4](#) [Q5](#) [Q6](#) [Q7](#) [Q8](#) [Q9](#) [Q10](#)